AI-Driven Antimicrobial Resistance (AMR) Prediction from Bacterial Genomes

Target Organism: *Klebsiella pneumoniae*
Target Antibiotic: Meropenem (Carbapenem class)

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1. The Core Problem: Predicting AMR from Genomes

- **What it is:** A machine learning pipeline trained to analyze the raw DNA sequence of a *Klebsiella pneumoniae* isolate and predict its phenotypic resistance (Susceptible vs. Resistant) to Meropenem.
- **Why it matters:** Traditional Antimicrobial Susceptibility Testing (AST) requires culturing bacteria in a lab with the antibiotic, taking 24–72 hours. In cases of sepsis, every hour of delayed targeted treatment increases mortality. Genomic sequencing is becoming rapid and inexpensive; an AI model that predicts resistance directly from DNA can guide life-saving clinical decisions on day one, bypassing the culture bottleneck.

2. Feature Extraction: Tokenization via K-mers

- **What it is:** Tokenizing the 5.5-million-character continuous string of DNA (`A, C, T, G`) into overlapping substrings of 31 characters (31-mers) using the KMC algorithm.
- **Why we use this approach:**
  - **Alignment-Free:** Resistance genes often reside on mobile genetic elements (plasmids) that jump between bacteria. K-mer counting ignores structural rearrangements and simply detects the presence or absence of resistance signatures anywhere in the genome.
  - **Computational Ef ciency (Why 31?):** A DNA base can be encoded in 2 bits ($2^2 = 4$ bases). Therefore, a 31-mer sequence takes exactly 62 bits, f tting perfectly into a standard 64-bit CPU register. This allows for lightning-fast bitwise operations during matrix construction.

3. Data Engineering: Preventing Clonal Leakage

- **What it is:** Using the Mash algorithm (MinHash sketching) to calculate pairwise genetic distances between all genomes, clustering them by >95% Average Nucleotide Identity (ANI), and performing a Grouped Train/Test Split based on Cluster IDs.
- **Why it is critical:** Bacteria replicate clonally. A hospital outbreak might contribute 50 nearly identical genomes to the dataset. A standard random 80/20 split would place

clones in both the training and testing sets, causing massive data leakage. The model would memorize the genetic background of specific outbreaks rather than learning the actual biological mechanisms of resistance. Grouping forces the model to generalize across entirely unseen lineages.

4. Data Architecture: SciPy Sparse Matrices

- **What it is:** Storing the final dataset (Genome Samples $\times$ K-mer Counts) as a Compressed Sparse Row (CSR) matrix rather than a dense tabular array (like a Pandas DataFrame).
- **Why it is required:** The dataset contains 4,300+ genomes and roughly 10 million unique 31-mer features. A dense 32-bit integer matrix of this size requires ~170 GB of RAM, which guarantees a system crash. Because a single genome contains only a tiny fraction of the global 10-million k-mer vocabulary, the matrix is >99% zeros. Sparse matrices only store non-zero values, reducing memory consumption to a few gigabytes.

5. The ML Algorithm: LightGBM / XGBoost

- **What it is:** Utilizing Gradient Boosted Decision Trees (GBDTs) for the classification task.
- **Why we chose it over Deep Learning:**
  - **The Curse of Dimensionality:** Standard neural networks instantly overfit when features (10 million) so vastly outnumber samples (4,300).
  - **Sparsity Handling:** Tree-based algorithms naturally excel at navigating extreme sparsity, only branching on features that are present.
  - **Class Imbalance:** Clinical resistance data is inherently skewed (e.g., 85% Susceptible, 15% Resistant). Gradient boosting natively balances this via the `scale_pos_weight` parameter, adjusting the loss function to heavily penalize false negatives (missing a resistant strain).

6. Biological Interpretability: SHAP and BLAST

- **What it is:** Extracting the most predictive features using SHAP (SHapley Additive exPlanations), converting those integer-based features back into 31-character DNA strings, and aligning them against the Comprehensive Antibiotic Resistance Database (CARD) using BLAST.
- **Why it validates the project:** In healthcare, black-box models are unacceptable. SHAP mathematically isolates exactly which k-mers drove a "Resistant" prediction. By confirming via BLAST that these mathematical tokens perfectly match known, real-world biological resistance genes (e.g., *blaNDM-1* or *blaKPC*), we prove the AI successfully reverse-engineered molecular biology without human intervention